Savitribai Phule Pune University, Pune

S.Y.B.Sc. (Computer Science)

Mathematics Practical Examination

MTC-292 MN-P: Practical on Mathematics for Computer Science-II

(NEP 2024 Pattern) (Semester-IV)

Slip No.:-**1**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Represent the following information using a **bar graph (in brown colour)** ,

Subject	Mathematics	English	Biology	Physics	Chemistry
Marks	68	45	79	56	70

2. Plot the graph of $y = x^3 + 10x - 5$, for $x \in [-10, 10]$ in red colour with dashed lines.

3. Draw surface plot for the function $f(x, y) = x^2 + y^2$ when $-10 \le x, y \le 10$, with pink colour.

4. Using sympy declare points P(7,2) , Q(1,8). Find the distance between them. Declare the line passing through them. Find the slope and equation of the line passing through points P and Q.

5. Using sympy declare Δ ABC where A (2 , 2), B(4, 2), C(3, 6) ,for this triangle find its centroid and also translate Δ ABC in x direction by 2 units.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the polygon with vertices at P(4 , 3),Q(6, 3),R(6, 5), and S(4, 5). Plot its reflections through, 1) x-axis, 2) y-axis . Also plot x axis and y axis in black colours . All figures must be in different colours .Use labels and legend .

2. Write a python program to generate and plot 100 points on circle $x^2+y^2 = 25$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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Slip No.:-**2**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Represent the following information using a **Pie chart** with percentage.

Sport	Cricket	Football	Badminton	Hockey	Other
Number of students	34	50	24	10	82

2. Plot the graph of $y = x^2 + 5$, for x ∈ [–5, 5] in green colour with dashed lines.

3. Draw Wireframes plot for the function $f(x, y) = (x^2 - y^2)$ when $-20 \le x, y \le 20$, with brown colour.

4. Using sympy declare point P(–4,5), apply each of the given transformation on point P.

a) Scaling in x co-ordinate by 2 unit, **b)** reflection through the line y = x, **c)** translation in y direction by factor 7 units.

5. Using sympy declare Δ ABC where A (3 , 0), B(0, 4), C(–3, 0) , for this triangle find its area, perimeter and all angles.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the polygon with vertices A(1, 1), B(–5, 1), C(–5, – 4) and D(1, – 4). On this polygon apply **1)** rotation about origin through angle 90^0, **2)** shearing in x-direction proportional to y by factor 3.

Plot transformed polygons also. All figures must be in different colours. Use legend.

2. Write a python program to generate and plot 100 points on circle $x^2+y^2 = 49$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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(NEP 2024 Pattern) (Semester-IV)

Slip No.:-**3**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Represent the following information using a **horizontal bar graph (in brown color)** ,

Subject	Mathematics	English	Biology	Physics	Chemistry
Marks	45	35	85	50	60

2. Plot the graph of $y = x^3 - 3x + 2$, for x ∈ [–5, 5] in maroon colour with arrow lines.

3. Draw surface plot for the function $f(x, y) = \sqrt{x^2 + y^2}$ when –20 ≤x, y≤ 20, with red colour.

4. Using sympy declare points L(–5,0) , M(4,0) and N(9,0). Determine whether the points L, M and N are collinear. Declare the line passing through the point L and N, find the slope and equation of the line passing through points L and N.

5. Using sympy declare regular polygon with center at origin, radius 2 and number of sides 5. Find its area, perimeter, centroid and also translate it in y direction by 4 units.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the polygon with vertices at points A(0,0), B(1, 0), C(1, 1) and D (0 , 1), find the transformed polygon by applying combined transformations using sequence **a)** reflections thorugh x-axis, **b)** rotation about origin through angle 45^0 and **c)** uniform scaling by factor -3 units. Both the figures must be in different colours.

2. Write a python program to generate and plot 50 points on parabola $y^2 = 12x$, in the where $0 \leq x \leq 20$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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MTC-292MN-P: Practical on Mathematics for Computer Science-II

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Slip No.:-**4**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Write python program to generate and plot graph of function $f(x) = x^3$ for $x \in [-5,5]$ and $g(x) = e^x$ for $x \in [1,10]$ in one graph window using 2 ×1 subplots.

2. Plot the graph of $y = 2x + 5$, for x ∈ [1, 20] in sky blue color with dashed lines.

3. Draw Wireframes plot for the function f(x, y) = cos (x^2+y^2) when 0 ≤x, y≤ 2π, with brown colour.

4. Using sympy declare the unit square. Find its area, perimeter and centroid.

5. Using sympy declare 3D Point P(2 , –1 , 3) and rotate about z-axis through angle 90^0 ,

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the Δ ABC with vertices A(–2 , 7), B(–5 , –1) and C(4, 3) On this polygon apply **1)** shearing in x-direction proportional to y by factor 3 and y-direction proportional to x by factor 4, **2)** reflection through x-axis, **3)** translation in y direction by factor 7 units. Plot original and transformed figures in different colour .

2. Write a python program to generate and plot 75 points on circle $x^2 + y^2 = 64$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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Slip No.:-**5**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Represent the following information using a **bar graph (All bars with different colours)** ,

Year	2016	2017	2018	2019	2020
Sale of Newspaper	45	50	60	55	70

2. Plot the graph of $y = x^2 + 2x + 5$, for $x \in [-20, 20]$ in red colour with dashed lines.

3. Draw surface plot for the function $f(x, y) = \sqrt{x^2 - y^2}$ when $-30 \le x, y \le 30$, with brown colour.

4. Using sympy declare line AB passing through points A(2, –4), B(2,8) and line PQ passing through point P(5,4), Q(–8,3). Find point of intersection of lines AB and PQ, also equations of Lines AB and PQ, and distance between point A and line PQ.

5. Using sympy declare triangle OAB where O (0, 0), A(7, –2), B(3, 6) , for this triangle find the scaling in x and y co-ordinates by factors 3 and –7 respectively and also find its reflection through the line y = x.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the 3d line segment with vertices at A(1, 2, 3), B(4 , 5 , 6). Find and plot **a)** reflection of AB through XY-plane, **b)** shearing of AB in y co-ordinate proportional to z co-ordinate by factor 4.

2. Write a python program to generate and plot 50 points on the parabola with equation $y^2 = 8x$ where $0 \le x \le 30$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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Slip No.:-**6**

Time: 3 hours | Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Write python program to plot function $f(x) = x^2$ for $x \in [-10,10]$ and $g(x) = e^{2x+4}$ for $x \in [1,20]$ in one graph window by using 1×2 subplots.

2. Plot the graph of $y = x^2 + 5$, for $x \in [-10,10]$ in red color using point marker arrow.

3. Draw Wireframes plot for the function $f(x, y) = \sin(x^2+y^2)$ when $0 \le x, y \le 2\pi$, with brown colour.

4. Using sympy declare the regular polygon with center at origin, radius 4 and number of sides 5. Find its area and also find its scaling in x and y directions by factors 3 and 2 respectively.

5. Using sympy declare polygon ABCD with vertices A(2, 2), B(6, 2), C(6, 6) and D(2, 6). Find its area, perimeter and all angles.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the Δ ABC with vertices A(1, 6), B(0 , 8) and C(5 , 3). Find and its **1)** shearing in x-direction proportional to y by factor 5, **2)** reflection through y-axis , **3)** translation x and y direction by factor 1 and 4 units respectively. And plot all transformed figures in different colours.

2. Write a Python program to generate and plot 75 points on circle $x^2+y^2 = 36$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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MTC-292MN-P: Practical on Mathematics for Computer Science-II

(NEP 2024 Pattern) (Semester-IV)

Slip No.:- **7**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Represent the following information using a **Pie chart.** Use explode function for separate the sectors.

Items	Food	Education	Rent	Travelling	Others
Expenses (in Thousand Rs.)	20	40	30	10	15

2. Plot the graph of $y = e^x$, for x ∈ [0, 10] in any color and dotted pattern .

3. Draw Surface plot for the function f(x, y) = $e^{(x^2+y^2)}$ when $-2\pi \le x, y \le 2\pi$, with red colour.

4. Using sympy declare line segment passing through points P(4, –5) and Q(–7,6). Apply each of the given transformation on segment PQ. **a)** reflection through the line y = –x,

b) Translation in x and y direction by factor 2 units and -3 units respectively.

5. Using sympy declare polygon with vertices A (4 , 3), B(–4, 3), C(– 4, –3) and D(4, –3) ,for this polygon find its area, perimeter, centroid and all angles.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the Pyramid OABC with the vertices O(0, 0, 0), A(1, 1, 0), B(0, 1, 0) and C(0, 0, 1). Find and plot its **1)** rotation about x-axis through angle 90^0,

2) reflection through YZ-plane .

2. Write a Python program to generate and plot 50 points on parabola with equation $y^2 = 20x$, where $0 \le x \le 40$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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Slip No.:- **8**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Represent the following information using a **bar graph .**

City	Satara	Dhule	Nashik	Nagpur	Pune
Temperature in ℃	20	32	25	40	30

2. Plot the graph of $y = \cos(x) + \sin(x)$, for x ∈ $[-\pi, \pi]$ in maroon colour with dashed lines.

3. Draw Wireframe plot for the function $f(x, y) = \sin(\sqrt{x^2 + y^2})$ when $0 \leq x, y \leq 2\pi$, with brown colour.

4. Using sympy declare points L(2, 6) , M(–2, 9) and N(–3, –5). Determine whether the points L, M and N are collinear, find distance between L and M, distance between N and M and distance between L and N and slope of line passing through Point L and M.

5. Using sympy declare regular polygon with center at origin, radius 6 and number of sides 6. Find its area, perimeter and translate this regular polygon in x direction by 8 units.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the polygon with vertices A(–2 , 5), B(–2, 9) , C(3 , 5) and D(4, 2) . Apply following transformations on polygon ABCD and plot the transformed polygons also, **a)** translation in x and y direction by factor 2 and –5 respectively, **b)** rotation about origin through angle 60^0 and **c)** shearing in x direction by factor 5 units.

2. Write a python program to generate and plot 100 points on circle $x^2 + y^2 = 81$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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MTC-292MN-P: Practical on Mathematics for Computer Science-II

(NEP 2024 Pattern) (Semester-IV)

Slip No.:- **9**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Write python program to plot the graph of function $f(x) = sin\ (x^2)$ for $x \in [-10,10]$ and $g(x) = \cos\ (x^2)$ for $x \in [0, \pi]$ in one graph window 2×1 subplots.

2. Plot the graph of $y = x^2 + e^{2x+5}$, for x ∈ [–10,10] in red color with dashed dotted line.

3. Draw Surface plot for the function f(x, y) = $x^3 + y^3$, when –20 ≤x, y≤ 20, with pink colour.

4. Using sympy declare unit square at origin and find its area, perimeter, centroid and all angles.

5. Using sympy declare 3D line segment passing through A(2 , 3, 5) and B(– 4, 7,9). And reflect through XZ-plane.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the polygon ABCD with vertices A(4, 4), B(–4, 4), C(–4, –4) and D(4 , –4). Apply the following sequence of combined transformations on it

1) shearing in y-direction by factor –3, **2)** reflection through x-axis, **3)** translation x and y direction by factors 5 and 8 respectively. And plot transformed polygon in different colours.

2. Write a python program to generate and plot 75 points on parabola $y^2 = 36x$ where $0 \leq x \leq 30$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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Slip No.:-**10**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Represent the following information using a **horizontal bar graph** (in blue color) ,

Year	2020	2021	2022	2023	2024
Number of Matches	50	40	30	67	45

2. Plot the graph of $y = x^2 - 5$, for $x \in [-10, 20]$ in red colour with lines using arrow marker.

3. Draw Wireframe plot for the function $f(x, y) = cos\sqrt{x^2 - y^2}$ when $0 \leq x, y \leq 2\pi$, with green colour.

4. Using sympy declare a regular polygon with center at origin, radius 5 and number of sides 4. Find its area, perimeter, vertices and all angles.

5. Using sympy declare 3D line with vertices at A(2, 0, 5), B(1, 5 , 8). And Rotate about Y-axis through an angle π.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the triangle OAB where O (0, 0), A(8, –9), B(7, 4) ,for this triangle find **a)** scaling in x and y co-ordinate by factor 5 and–6 respectively, **b)** reflection through x axis and **c)** rotation about origin through angle 60^0. And plot all transformed figures in different colours.

2. Write a python program to generate and plot100 points on circle $x^2 + y^2 = 144$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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Slip No.:-**11**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4= 20 Marks]**

1. Draw the **Histograph** with given data in brown colour,which will give us the number of students obtaining the marks between 20– 40 ,40– 60 , 60 – 80 and 80–100
   Marks of students: 71,29,35,21,66, 32, 24,27,38,44,51,42,49,86,94,46,57,47,41,50,46.

2. Plot the graph of $y = x^2 + \cos(2x + 5)$, for x $\in [0, \frac{\pi}{2}]$ in red colour with arrow lines.

3. Draw Surface plot for the function $f(x, y) = Sin\sqrt{x^2 + y^2}$ when 0 ≤x, y≤ 2π, with brown colour.

4. Using sympy generate the lines with equations 3 x+ 5y =4 and x – 5y =8. Find point of intersection of these two lines and angle between these two lines.

5. Using sympy declare polygon with vertices O(0, 0), A(5, 0), C(5, 2) and D(0, 2). Determine whether point (5,1) is inside the polygon. Also find the translation of polygon in x and y direction by factor 3 and 3 units respectively.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to generate and plot the Δ ABC, where A(10, 5), B(–7, 4) and C(8, –3). Find and plot the transformed triangle by applying combined transformations of sequence **a)** reflection through y-axis, **b)** rotation about origin through angle 90^0 and **c)** uniform scaling by factor 7 units.

2. Write a Python program to generate and plot 50 points on parabola $y^2 = 40x$ for $0 \leq x \leq 50$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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Slip No.:- 12

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4 = 20 Marks]**

1. Represent the following information using a **Pie Chart**

Subject	Mathematics	English	Biology	Physics	Chemistry
Marks	84	48	60	35	72

2. Plot the graph of $y = \cos(2x)$, for $x \in [-\pi, \pi]$ in red colour with dashed lines.

3. Draw wireframe plot for the function $f(x, y) = \sin(x^2 + y^2)$ when $-10 \le x, y \le 10$, with brown colour.

4. Using sympy declare points P (0, 0) , Q (1 , 1), R (– 7 ,5) . Find the distance of points Q and R from P. Also find midpoint of line segment QR.

5. Using sympy declare polygon ABCD where A (1 , 1), B (–1, 1), C (–1, –1) and D (1 , –1) , Find angles of polygon ABCD at each vertices. Also find its rotation about origin by $\frac{\pi}{3}$ radians.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the Triangle with vertices at P (1 , 3), Q (3 , 3), R(4 , 5) and its reflections through, 1) line y = x 2) line y = – x . All the figures must be in different colours.

2. Write a python program to generate and plot 50 points on circle $x^2 + y^2 = 16$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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Slip No.:- 13

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4 = 20 Marks]**

1. Represent the following information using a **Pie Chart showing percentage.**

Subject	Mathematics	English	Biology	Physics	Chemistry
Marks	85	15	50	67	77

2. Plot the graph of $y = \cos(x^2)$, for $x \in [-\pi, \pi]$ in red colour with dashed lines.

3. Draw wireframe plot for the function $f(x, y) = (x^2 - y^2)$ when $-10 \le x, y \le 10$, with brown colour.

4. Using sympy declare points P (0, 0) , Q (1 , 1), R (–1 ,1) , S (–1, –1), T (1 , –1) , Rotate the polygon about origin through π radians also find centroid of polygon.

5. Using sympy declare the Triangle with vertices at A[1 3], B[3 3], C[4 5] . Find angles of Δ ABC at each vertices , also find its area and perimeter.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the Polygon ABCD where A (0 , 3), B (–3, 3), C (–3, 0) and D (0 ,0). Find and plot, **1)** reflection of ABCD through line y = x , **2)** Uniform Scaling of ABCD by facor 4. All the figures must be in different colours.

2. Write a Python program to generate and plot 50 points on circle $x^2 + y^2 = 25$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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Slip No.:- 14

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4 = 20 Marks]**

1. Represent the following information using a **bar graph (in brown color)** ,

Subject	Mathematics	English	Biology	Physics	Chemistry
Marks	68	45	79	56	70

2. Plot the graph of $y = \log(x)$, for $x \in (0 , 10]$ in red colour with dashed lines.

3. Draw surface plot for the function $f(x, y) = (x^3 + y^3)$ when $-10 \leq x, y \leq 10$, with green colour.

4. Using sympy declare points P (7,2) , Q (1,8). Find the distance between them. Declare the line passing through them. Find the slope and equation of the line passing through points P and Q.

5. Using sympy declare Δ ABC where A (2 , 2), B (4, 2), C (3, 6) ,for this triangle find its centroid and also translate Δ ABC in x direction by 2 units.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the polygon with vertices at (1, 3), (2, 4), (4 , 2) and (3,1) Plot its rotation about ,origin through angles 1) $\frac{\pi}{2}$ radians and **2)** π radians .

Also plot x and y axis in black colours.

2. Write a Python program to generate and plot 50 points on parabola $y^2 = 4x$ with $4 \leq x \leq 16$ using iterative formulae.

Q.3 Viva Voce **[5 Marks]**

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(NEP 2024 Pattern) (Semester-IV)

Slip No.:- **15**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4 = 20 Marks]**

1. This is information of students participating in various games, represent it by a horizontal Bar graph.

Game	Cricket	Football	Hockey	Chess	Tennis
Number of students	125	59	28	19	36

2. Plot the graph of $f(x) = \sin(x) + \cos(x)$, for $x \in [-3\pi, 3\pi]$ in dashed dotted pattern.

3. Plot 3D surface plot for the function $f(x, y) = x^4 - y^2$ for $-5 \le x, y \le 5$, in black .

4. Using sympy declare the points P(7, 3), Q(7,−5) and R(7,1), check whether these points are collinear. Declare the segment passing the points P and Q, find the midpoint of segment PQ.

5. Using sympy declare Δ ABC where A (2 , 2), B(4, 2), C(3, 6) ,for this triangle find its centroid and also translate ΔABC in x direction by 2 units.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to perform each of the following transformations on the point (5, −8). Plot the point and its following transformations using different colors and markers.

a) Refection through Y−axis, **b)** Scaling in x−coordinate by factor 4.

c) Rotation about origin through an angle $\frac{\pi}{2}$.

2. Write python program to generate and plot 50 points on parabola $y^2 = 20x$, using iterative formulae for $0 \le y \le 30$.

Q.3. Viva Voce. **[5 Marks]**

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(NEP 2024 Pattern) (Semester-IV)

Slip No.:- **16**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4 = 20 Marks]**

1. Show the following information by **pie graph**. Use labels and display percentage.

Name of the Fruit	Mango	Grapes	Banana	Other
Number of students who like it	65	30	5	20

2. Plot straight line joining the points [2 4] and [3 6] in green colour.

3. Draw 3D wireframe plot of the function $f(x,y) = -x^3 - y^3$ in Brown color for, $x, y \in [-15, 15]$

4. Using sympy declare quadrilateral ABCD with vertices A (6,2) , B (4,8), C(12,9) and D(5,3). Find the area and perimeter of □ABCD.

5. Using sympy declare Δ PQR where P (1 , 1), Q(5, 1), R(5, 5) ,for this triangle find its area, circumcircle, Is it right angled triangle ? Scale Δ PQR in x and y directions by factors 2 and 4 respectively .

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the polygon with vertices at points (1, 2) , (4 2), (4 4) and (2,5). Plot shearing of this polygon shearing in x and y direction by 3 and -2 units respectively. Also plot x axis and y axis in black colours.

2. Write a python program to generate and plot 80 points on parabola $y^2 = 16x$ using iterative formulae for $0 \le x \le 80$, .

Q.3. Viva Voce . **[5 Marks]**

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Savitribai Phule Pune University, Pune

S.Y.B.Sc. (Computer Science)

Mathematics Practical Examination

MTC-292 MN-P: Practical on Mathematics for Computer Science-II

(NEP 2024 Pattern) (Semester-IV)

Slip No.:- **17**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4 = 20 Marks]**

1. Represent the following information using a **pie chart** , by using explode function to separate sectors.

Subject	Computer	Mathematics	Electronics	Statistics	English
Marks	60	80	65	55	70

2. Plot the graph of $f(x) = \sin(3x + 1)$, for $x \in [-5\pi, 5\pi]$ in red colour and dotted pattern.

3. Draw surface plot for the function $f(x, y) = -(x^2 + y^2)$ for $x, y \in [-10, 20]$.

4. Using sympy find the point of intersection of two lines L_1 and L_2, given by the equations $x + y = 2$ and $3x+2y = 1$. Also find their slopes.

5. Using sympy declare a regular polygon with 6 sides and radius 1 centered at (1, 2) and find its area perimeter and centroid.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Using python program find the **combined transformation** of the line segment joining the points P[5,−2] and Q[3, 2] for the following sequence of transformations:-

(a) Rotation about origin through an angle π . **(b)** Scaling in X−coordinate by 4 units.

(c) Reflection through the line y = x. Also plot original and transformed line segments.

2. Write python program to generate and plot100 points on circle $x^2 + y^2 = 49$ using iterative formulae.

Q.3. Viva Voce . **[5 Marks]**

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Savitribai Phule Pune University, Pune

S.Y.B.Sc. (Computer Science)

Mathematics Practical Examination

MTC-292 MN-P: Practical on Mathematics for Computer Science-II

(NEP 2024 Pattern) (Semester-IV)

Slip No.:- **18**

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4 = 20 Marks]**

1. The age of 25 persons who donated blood in camp arranged for the age group 20 to 45 yrs is given below. Plot the histograph in Maroon colour which will give us the number of donors in the age groups 20-25,25-30,30-35 ,35-40 and 40-45

   Ages are: 21, 29, 35, 21, 41, 36, 32, 24, 43,27,38,34,31,22,39,26,34,26,44,37,37,31,30,22,39.

2. Plot the graph of $y = 2x^2 + 6x + 10$, for $-7 \le x \le 10$.

3. Draw surface plot for the function $f(x, y) = (-x^2 - y^2)$ for $x, y \in [-15, 10]$, in Green colour.

4. Using sympy declare two lines L_1 and L_2 given by the equations $x + y = 2$ and $3x+2y = 1$. Find the point of intersection of lines L_1 and L_2. Also find the angle between these two lines.

5. Using sympy declare polygon P with vertices A(1 , 3), B(3, 5), C(5 , 4), D (4, 2), Determine whether it is convex polygon? Rotate polygon P about point (2,3) by angle $\frac{\pi}{3}$.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to generate and plot the Δ PQR with vertices at P (1, 2), Q(4 2), R(4, 5). And also plot its uniform scaling by factor 0.5 and uniform scaling by factor 4.

   All three figures must be in different colours.

2. Write python program to generate and plot 60 on the circle with equation $x^2 + y^2 = 100$ using iterative formulae.

Q.3. Viva Voce . **[5 Marks]**

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Savitribai Phule Pune University, Pune

S.Y.B.Sc. (Computer Science)

Mathematics Practical Examination

MTC-292 MN-P: Practical on Mathematics for Computer Science-II

(NEP 2024 Pattern) (Semester-IV)

Slip No.:-19

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4 = 20 Marks]**

1. Represent the following information using pie chart with detached sectors .

Wild Animals	Grazing Animals	Birds	Water Animals	Reptiles
150	400	225	175	50

2. Plot 50 points on the curve y= tan(x) in $[0, \frac{\pi}{2}]$ in green colour.

3. Plot Surface plot for the function $f(x, y) = -x^3 - y^3$, where $x, y \in [1, 10]$.

4. Using sympy find the angle between the line segment AB joining the points A[7,2], B[3,−4] and the line having equation $2x - y = 1$.

5. Using sympy declare polygon ABCD with vertices A(3, 3), B(3, 6), C(8, 6) and D(7, 2) ,for this polygon determine whether point (4,4) inside . And scale ABCD in x and y directions by factors 4 and 3 respectively.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to find the transformation of the line segment joining the points A(1,−2) and B(3,5) for the following sequence of transformations:

(I) Rotation about origin through an angle π. **(II)** Scaling in X− coordinate by 7 units.

(III) Uniform scaling by factor 4. Plot original and transformed line segment.

2.Write python program to generate and plot 50 points on parabola $y^2 = 28x$, using iterative formulae for $0 \le x \le 100$.

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Savitribai Phule Pune University, Pune

S.Y.B.Sc. (Computer Science)

Mathematics Practical Examination

MTC-292 MN-P: Practical on Mathematics for Computer Science-II

(NEP 2024 Pattern) (Semester-I)

Slip No.:- 20

Time: 3 hours Maximum Marks: 35

Q.1. Attempt any ***four*** of the following. **[5×4 = 20 Marks]**

1. Draw a pie chart for the given data with radius 0.7 , show percentage :

Subject	History	Geography	English	Sociology	Psychology
Number of students	168	90	195	95	46

2. Plot the graph of $x^2 + 5x$ in [5, 10] with dotted pattern.
3. Using sympy declare points A (1 ,1) , B(1, 6), C (6 ,1) , D(6, 6) and plot quadrilateral ABCD in black color.
4. Using sympy declare quadrilateral ABCD where A (1 ,1), B(1, 4), C(4, 1) and D(4, 4), and find shearing of ABCD in x and y directions by 3 and 7 units respectively
5. If the line segment joining the points A[−3, 5] and B[3, 3] is transformed to the line segment A'B' by the transformation matrix [T] =$\begin{bmatrix} 1 & 3 \\ 4 & 1 \end{bmatrix}$,then using sympy find the midpoint of the transformed line.

Q.2. Attempt any ***one*** of the following. **[10 ×1= 10 Marks]**

1. Write a python program to plot the triangle with vertices at (4,4) , (2, 4) ,(4, 2) . Apply the transformations **a)** its rotation about the origin by $\frac{\pi}{2}$ radians, **b)** uniform scaling by factor 3. Plot original and transformed figures in different colours Also plot x and y axes in one colour.
2. Write Python program to generate and plot 100 points on parabola $y^2 = 32x$, using iterative formulae for $0 \le x \le 100$.

Q.3. Viva Voce . **[5 Marks]**

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